

Problem Statement — *House Price Prediction Using IBM SPSS Modeler*

The real estate market is rapidly growing, and determining the correct price of a house has become a challenging task due to the involvement of multiple factors such as location, area, number of bedrooms, bathrooms, furnishing condition, parking availability, and age of the property.

Traditional methods of estimating house prices are often time-consuming and may produce inaccurate results because they depend heavily on manual calculations and human judgment. Therefore, there is a need for an intelligent system that can automatically analyze housing data and predict accurate property prices.

The objective of this project is to develop a House Price Prediction System using IBM SPSS Modeler and machine learning techniques. The system uses historical housing data to train predictive models and estimate house prices based on important housing features. Various data preprocessing techniques such as filtering, deriving fields, partitioning data, and statistical analysis are performed to improve model accuracy. Machine learning algorithms like Linear Model, Regression, CHAID, and Neural Networks are used to identify relationships between house features and prices.

The developed system helps buyers, sellers, and real estate agencies make better decisions by providing fast and accurate house price predictions. This project demonstrates the practical application of predictive analytics and machine learning in solving real-world real estate problems efficiently.